

## KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6351B

6 Credits: 4

Programme Elective Course

Source-grounded

# Solid Waste Management and

□ To enable students to familiarize with various functional elements involved in Solid

## OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 6351B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 6351B.pdf from the uploaded official SITTTTR syllabus archive.

**Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

| Item                 | Official information                                 |
|----------------------|------------------------------------------------------|
| Programme            | Diploma in Civil and EnvironmentalEngineering        |
| Course code          | 6351B                                                |
| Course title         | Solid Waste Management and                           |
| Semester             | 6 Credits: 4                                         |
| Course category      | Programme Elective Course                            |
| Periods per week     | 4(L:4 T:0 P:0) Periods per semester:60               |
| Periods per semester | 60                                                   |
| Credits              | 4                                                    |
| Prerequisite         | See the official syllabus PDF                        |
| Assessment           | Use the official assessment section reproduced below |

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

**STUDY METHOD****How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

**Second pass**

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

**Practical evidence**

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

**Revision pass**

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

**LEARNING OUTCOMES****Objectives, outcomes and mapping****Course objectives**

- To enable students to familiarize with various functional elements involved in Solid
- Waste management.
- To enable students to discuss various methods for solid waste disposal.
- To enable students to identify impacts of air pollution and discuss various methods for
- air pollution control.

**Course outcomes**

| CO  | Official outcome                                                                                                                                             | Study level           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | and characteristics and identify steps in waste 14 collection, storage and transportation. Discuss various methods for disposal of solid waste Understanding | See official syllabus |
| CO2 | 15 based on waste characteristics. Identify sources, impacts and sampling and Understanding                                                                  | See official syllabus |
| CO3 | 15 measurement methods of air pollutants.                                                                                                                    | See official syllabus |
| CO4 | Discuss various methods for air pollution control. 14 Understanding Tests 2                                                                                  | See official syllabus |

**CO-PO mapping evidence**

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3 2 2 3                |
| CO2 | CO2 3 2 2 3                |
| CO3 | CO3 3 2 2 3                |

| Row | Extracted official mapping |
|-----|----------------------------|
| CO4 | CO4 3 2 2 3                |

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

| Module   | Official heading                                   | Topic entries | Hours        |
|----------|----------------------------------------------------|---------------|--------------|
| Module 1 | in waste collection, storage and transportation.   | 4             | As specified |
| Module 2 | characteristics.                                   | 3             | As specified |
| Module 3 | pollutants.                                        | 4             | As specified |
| Module 4 | Discuss various methods for air pollution control. | 4             | As specified |

**MODULE 1**

# in waste collection, storage and transportation.

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: in waste collection, storage and transportation.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ Classify solid waste based on source. 1 Understanding Describe characteristics of each category of

Classify solid waste based on source. 1 Understanding Describe characteristics of each category of is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Classify solid waste based on source. 1 Understanding Describe characteristics of each category of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, classify solid waste based on source. 1 understanding describe characteristics of each category of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Classify solid waste based on source. 1 Understanding Describe characteristics of each category of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-  
 Classify solid waste based on source. 1  
 Understanding Describe characteristics of each category of   
 definition/meaning .   
 , steps, application . Technical terms English-  
 .



## ▼ 5 Understanding waste. Identify purpose and functional elements of solid

5 Understanding waste. Identify purpose and functional elements of solid is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding waste. Identify purpose and functional elements of solid using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding waste. identify purpose and functional elements of solid should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding waste. Identify purpose and functional elements of solid means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-5 Understanding waste. Identify purpose and functional elements of solid waste. definition/meaning, components, steps, application, examples. Technical terms English-Malayalam.

### ▼ 3 Understanding waste management. Describe waste collection, storage and

3 Understanding waste management. Describe waste collection, storage and is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding waste management. Describe waste collection, storage and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding waste management. describe waste collection, storage and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding waste management. Describe waste collection, storage and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-3 Understanding waste management. Describe waste collection, storage and definition/meaning. Steps, application. Technical terms English-  
.

**▼ 5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste**

5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal,Industrial, Bio-medical, Universal, Construction and demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components,storagecontainers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction,chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste is an official topic in Module 1 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

## Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

## Study points

- Define or identify 5 Understanding transportation system Total 14  
Introduction to the course Wastes-Classification based on source-  
Municipal,Industrial, Bio-medical, Universal, Construction and  
demolition,Radioactive, e wastes, Agricultural waste- Characteristics of each  
type, necessity of safe disposal of solid wastes. Hazardous waste-  
Identification and classification of hazardous waste. Solid waste  
management- Purpose, Functional elements, hierarchy of waste  
management options. Solid Waste management rules 2016 Waste  
Collection, Storage and Transport: Collection  
components,storagecontainers/collection vehicles, collection operation,  
transfer station, waste collection system design, record keeping, control,  
inventory and monitoring, implementing collection and transfer system.  
Processing techniques- Mechanical volume and size reduction,chemical  
volume reduction, component separation. Discuss various methods for  
disposal of solid waste based on waste using standard technical  
terminology.
- Arrange the important components, steps or characteristics in a logical  
sequence.
- Relate the topic to one laboratory, industrial, professional or design  
situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the  
question requires it.

## Engineering or professional relevance

In Polytechnic practice, 5 understanding transportation system total 14  
introduction to the course wastes-classification based on source-  
municipal,industrial, bio-medical, universal, construction and  
demolition,radioactive, e wastes, agricultural waste- characteristics of each type,

necessity of safe disposal of solid wastes. hazardous waste- identification and classification of hazardous waste. solid waste management- purpose, functional elements, hierarchy of waste management options. solid waste management rules 2016 waste collection, storage and transport: collection components, storage containers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. processing techniques- mechanical volume and size reduction, chemical volume reduction, component separation. discuss various methods for disposal of solid waste based on waste should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal, Industrial, Bio-medical, Universal, Construction and demolition, Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components, storage containers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction, chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- 5 Understanding transportation system  
Total 14 Introduction to the course Wastes-Classification based on source- Municipal, Industrial, Bio-medical, Universal, Construction and demolition, Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components, storage containers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction, chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste definition/meaning. steps, application. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.



**MODULE 2****characteristics.**

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

**Learning goals**

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

**Key facts**

- Official module title: characteristics.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## ▼ 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of

6 Understanding solid wastes. Discuss thermo chemical methods for disposal of is an official topic in Module 2 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 understanding solid wastes. discuss thermo chemical methods for disposal of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of   
 definition/meaning   
 application   
 Technical terms English-   
 .

## ▼ 6 Understanding solid wastes.

6 Understanding solid wastes. is an official topic in Module 2 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Understanding solid wastes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 understanding solid wastes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                         |
|---------------------|-------------------------------------------------------------------------|
| Definition/identity | What 6 Understanding solid wastes. means in the official course context |
| Study lens          | Principle → components or stages → result → application                 |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 6 Understanding solid wastes.

പ്രതിരോധം പ്രതിരോധം definition/meaning പ്രതിരോധം പ്രതിരോധം. പ്രതിരോധം പ്രതിരോധം, steps, application പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം. Technical terms English- പ്രതിരോധം പ്രതിരോധം പ്രതിരോധം.

▼ **Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting- Indore and Bangalore process Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air**

Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indore and Bangalore process Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods - incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air is an official topic in Module 2 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

## Study points

- Define or identify Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indoreand Bangalore process Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, discuss treatment methods for hazardous wastes 3 understanding series 1 1 total 16 contents sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. composting- stages in aerobic composting, types of composting-indoreand bangalore process anaerobic digesters – stages in anaerobic digestion, parts of a digester, products of anaerobic digestion, sludge disposal methods thermo chemical methods - incineration, gasification and pyrolysis. types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. hazardous waste management-handling, storage, collection, and transportation of hazardous waste. treatment and remedial actions- stabilization and solidification, thermal methods, secure landfill. identify sources, impacts and

sampling and measurement methods of air should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting-Indoreand Bangalore process Anaerobic digesters - Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas andcomposition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of



composting-Indore and Bangalore process Anaerobic digesters – Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air pollution. Definition/meaning of air pollution. Sources of air pollution, steps, application of air pollution control measures. Technical terms English-Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 3****pollutants.**

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

**Learning goals**

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

**Key facts**

- Official module title: pollutants.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ 6 Understanding pollutants. Identify effects of air pollutants of environment

6 Understanding pollutants. Identify effects of air pollutants of environment is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Understanding pollutants. Identify effects of air pollutants of environment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 understanding pollutants. identify effects of air pollutants of environment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Understanding pollutants. Identify effects of air pollutants of environment means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-6 Understanding pollutants. Identify effects of air pollutants of environment ഏകദേശം 6 Understanding pollutants. Identify effects of air pollutants of environment. definition/meaning. ഏകദേശം 6 Understanding pollutants. Identify effects of air pollutants of environment. steps, application. Technical terms English- Malayalam.

## ▼ 4 Understanding and health. Discuss sources and impacts of indoor air

4 Understanding and health. Discuss sources and impacts of indoor air is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding and health. Discuss sources and impacts of indoor air using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding and health. discuss sources and impacts of indoor air should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

**Malayalam support:** Module 3- 4 Understanding and health. Discuss sources and impacts of indoor air pollution. Define/meaning of indoor air pollution. Discuss the steps, application of indoor air pollution control measures. Technical terms English- Malayalam.

## ▼ 2 Understanding pollution. Explain air sampling and pollution measurement

2 Understanding pollution. Explain air sampling and pollution measurement is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding pollution. Explain air sampling and pollution measurement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding pollution. explain air sampling and pollution measurement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding pollution. Explain air sampling and pollution measurement means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 2 Understanding pollution. Explain air sampling and pollution measurement ഏകദേശം ൧൦൦ ശതമാനം definition/meaning എന്താണിത്. എങ്ങനെ അളക്കുന്നു steps, application എവിടെയും ഉപയോഗിക്കുന്നു. Technical terms English-ൽ പറയാൻ സഹായിക്കുക.



**▼ 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.**

3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. is an official topic in Module 3 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of

pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods- principles of sampling, source and ambient sampling, instruments used. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding methods. total 15 modern air pollutants- classification of pollutants, important gaseous pollutants and sources of pollutants. bio pollutants. effects of air pollutants on atmosphere- reactions of pollutants in the atmosphere and their effects- smoke, smog and ozone layer disturbance and greenhouse effect, acid rain. effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. indoor air pollution- sources, types and health impacts. air sampling and pollution measurement methods- principles of sampling, source and ambient sampling, instruments used. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. |

| Aspect        | What to prepare                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used. means in the official course context |
| Study lens    | Principle → components or stages → result → application                                                                                                                                                             |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 3 Understanding methods. Total 15  
 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.   
 definition/meaning   
 steps, application   
 Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 4**

# Discuss various methods for air pollution control.

This module is presented from the official syllabus scope for Solid Waste Management and. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Discuss various methods for air pollution control.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ 2 Understanding for air quality monitoring. Discuss various control methods for particulate

2 Understanding for air quality monitoring. Discuss various control methods for particulate is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding for air quality monitoring. Discuss various control methods for particulate using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding for air quality monitoring. discuss various control methods for particulate should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding for air quality monitoring. Discuss various control methods for particulate means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                               |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                     |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 2 Understanding for air quality monitoring. Discuss various control methods for particulate . definition/meaning . steps, application . Technical terms English- .

## ▼ 5 Understanding pollutants. Discuss various control methods for gaseous

5 Understanding pollutants. Discuss various control methods for gaseous is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding pollutants. Discuss various control methods for gaseous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding pollutants. discuss various control methods for gaseous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



| Aspect              | What to prepare                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding pollutants. Discuss various control methods for gaseous means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-5 Understanding pollutants. Discuss various control methods for gaseous pollutants. definition/meaning, components, steps, application. Technical terms English-Malayalam.

## ▼ 5 Understanding pollutants. Discuss various biological air pollution control

5 Understanding pollutants. Discuss various biological air pollution control is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding pollutants. Discuss various biological air pollution control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding pollutants. discuss various biological air pollution control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding pollutants. Discuss various biological air pollution control means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-5 Understanding pollutants. Discuss various biological air pollution control ജൈവ വായു മലിനീകരണ നിയന്ത്രണ definition/meaning അർത്ഥം. പ്രയോഗം പ്രയോഗം പ്രയോഗം, steps, application പ്രയോഗം പ്രയോഗം പ്രയോഗം. Technical terms English-ജൈവ വായു മലിനീകരണ നിയന്ത്രണ.

**▼ 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/>**

2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control

Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> is an official topic in Module 4 of Solid Waste Management and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air

pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 2 understanding techniques. series 2 1 total 15 ambient air quality and emission standards in india, air pollution indices, air act legislations and regulations. removal of gaseous pollutants by adsorption, absorption, condensation and combustion- principle, working, advantages and disadvantages of each method. particulate emission control by settling chambers, cyclone separation, wet collectors, fabric filters, electrostatic precipitators. principle, working, advantages and disadvantages of each method. biological air pollution control technologies. -biofilters, bio trickling filters and bio scrubbers. text / reference: t/r book title/author garg, s.k., “environmental engineering vol. ii”, khanna publishers, new delhi t1 , latest edition c. s rao, “environmental pollution control engineering” , new age t2 international publishers, 2006 t3 r c gaur, “basic environmental engineering”, new age publications, 2008 m. n. rao, h. v. n. rao, “air pollution”, tata mcgraw hill pvt. ltd, new delhi r1 2017 cpheeo. manual on municipal solid waste management. central public health r2 and environmental engineering organization, government of india, new delhi., 2016 peavy s.w., rowe d.r. and tchobanoglous g, “ environmental engineering” , r3 mcgraw hill, new delhi, 1985. keshav kant, “ air pollution control engineering”, khanna publishing house, r4 2023 online resources: sl.no. website link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> should be

discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | <p>What 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers.</p> <p>Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4</p> <p>2023 Online resources: Sl.No. Website Link 1 <a href="https://cpcb.nic.in/">https://cpcb.nic.in/</a> 2 <a href="https://nptel.ac.in/">https://nptel.ac.in/</a> means in the official course context</p> |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017 CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill, New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: Sl.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/> 3 definition/meaning 4 . 5 , steps, application 6 . Technical terms English- 7 .

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.



**RAPID REFERENCE****Concept and formula bank****Answer structure**

Definition → Principle →  
Components/steps →  
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

**Practical record**

Aim → Apparatus → Procedure  
→ Observation → Result →  
Safety

Use this sequence for laboratory, workshop and field activities.

**RAPID REVISION****Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 2 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 3 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 4 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**OFFICIAL SELF-LEARNING****Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

**EXAMINATION PREPARATION**

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

**ACTIVE RECALL****Module-wise questions and answers**

These are practice questions based on official topic coverage, not official examination questions.

**Module 1**

► **Define or explain Classify solid waste based on source. 1 Understanding Describe characteristics of each category of. (3 marks)**

► **Define or explain 5 Understanding waste. Identify purpose and functional elements of solid. (3 marks)**

► **Define or explain 3 Understanding waste management. Describe waste collection, storage and. (3 marks)**

► **Define or explain 5 Understanding transportation system Total 14 Introduction to the course Wastes-Classification based on source- Municipal, Industrial, Bio-medical, Universal, Construction and demolition, Radioactive, e wastes, Agricultural waste- Characteristics of each type, necessity of safe disposal of solid wastes. Hazardous waste- Identification and classification of hazardous waste. Solid waste management- Purpose, Functional elements, hierarchy of waste management options. Solid Waste management rules 2016 Waste Collection, Storage and Transport: Collection components, storage containers/collection vehicles, collection operation, transfer station, waste collection system design, record keeping, control, inventory and monitoring, implementing collection and transfer system. Processing techniques- Mechanical volume and size reduction, chemical volume reduction, component separation. Discuss various methods for disposal of solid waste based on waste. (3 marks)**

## Module 2

► Define or explain 6 Understanding solid wastes. Discuss thermo chemical methods for disposal of. (3 marks)

► Define or explain 6 Understanding solid wastes.. (3 marks)

► Define or explain Discuss treatment methods for hazardous wastes 3 Understanding Series 1 1 Total 16 Contents Sanitary landfills- parts and their functions, methods of landfilling, advantages and disadvantages. Composting- Stages in aerobic composting, types of composting- Indore and Bangalore process Anaerobic digesters - Stages in anaerobic digestion, Parts of a digester, Products of anaerobic digestion, sludge disposal methods Thermo chemical methods -incineration, gasification and pyrolysis. Types of incinerators - parts of an incinerator, effluent gas and composition, advantages and disadvantages. Hazardous waste management-Handling, Storage, Collection, and Transportation of hazardous waste. Treatment and remedial actions- Stabilization and Solidification, Thermal methods, Secure Landfill. Identify sources, impacts and sampling and measurement methods of air. (3 marks)

## Module 3

► Define or explain 6 Understanding pollutants. Identify effects of air pollutants of environment. (3 marks)

► Define or explain 4 Understanding and health. Discuss sources and impacts of indoor air. (3 marks)

► Define or explain 2 Understanding pollution. Explain air sampling and pollution measurement. (3 marks)

► Define or explain 3 Understanding methods. Total 15 Modern Air pollutants- Classification of pollutants, Important gaseous pollutants

**and sources of pollutants. Bio pollutants. Effects of air pollutants on atmosphere- Reactions of pollutants in the atmosphere and their effects- Smoke, smog and ozone layer disturbance and Greenhouse effect, Acid rain. Effects of air pollutants on human health, plants, live stock animals, materials and services, aquatic life, global and regional environment. Indoor air pollution- sources, types and health impacts. Air sampling and pollution measurement methods-principles of sampling, source and ambient sampling, instruments used.. (3 marks)**

## **Module 4**

► **Define or explain 2 Understanding for air quality monitoring. Discuss various control methods for particulate. (3 marks)**

► **Define or explain 5 Understanding pollutants. Discuss various control methods for gaseous. (3 marks)**

► **Define or explain 5 Understanding pollutants. Discuss various biological air pollution control. (3 marks)**

► **Define or explain 2 Understanding techniques. Series 2 1 Total 15 Ambient air quality and emission standards in India, Air pollution indices, Air Act legislations and regulations. Removal of gaseous pollutants by adsorption, absorption, condensation and combustion- Principle, working, advantages and disadvantages of each method. Particulate emission control by settling chambers, cyclone separation, Wet collectors, fabric filters, electrostatic precipitators. Principle, working, advantages and disadvantages of each method. Biological air pollution control technologies. -Biofilters, Bio trickling filters and bio scrubbers. Text / Reference: T/R Book Title/Author Garg, S.K., “Environmental Engineering Vol. II”, Khanna Publishers, New Delhi T1 , Latest Edition C. S Rao, “Environmental Pollution Control Engineering” , New Age T2 International Publishers, 2006 T3 R C Gaur, “Basic Environmental Engineering”, New Age Publications, 2008 M. N. Rao, H. V. N. Rao, “Air pollution”, Tata McGraw Hill Pvt. Ltd, New Delhi R1 2017**

**CPHEEO. Manual on Municipal solid waste management. Central public Health R2 and Environmental Engineering Organization, Government of India, New Delhi., 2016 Peavy S.W., Rowe D.R. and Tchobanoglous G, “ Environmental Engineering” , R3 McGraw Hill,New Delhi, 1985. Keshav Kant, “ Air Pollution Control Engineering”, Khanna Publishing House, R4 2023 Online resources: SI.No. Website Link 1 <https://cpcb.nic.in/> 2 <https://nptel.ac.in/>. (3 marks)**

**PRACTICE ONLY**

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Classify solid waste based on source. 1 Understanding Describe characteristics of each category of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **6 Understanding solid wastes. Discuss thermo chemical methods for disposal of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **6 Understanding pollutants. Identify effects of air pollutants of environment.**

### Module 4

1-mark definition: State the meaning of **2 Understanding for air quality monitoring. Discuss various control methods for particulate.**



3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

**REFERENCE VOCABULARY**

# Glossary

**Study glossary**

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

**SOURCE-SUPPORTED REFERENCES**

## References and web resources

**References listed in the official PDF**

References are included in the official source PDF.

**Web resources listed in the official PDF**

- Sl.No. Website Link <https://cpcb.nic.in/> <https://nptel.ac.in/>

**IN-PAGE SEARCH**

## Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 6351B. Verify institutional updates against the current official curriculum.